

SIMOG Operational Architecture

Deployment Constraints and Runtime Topologies

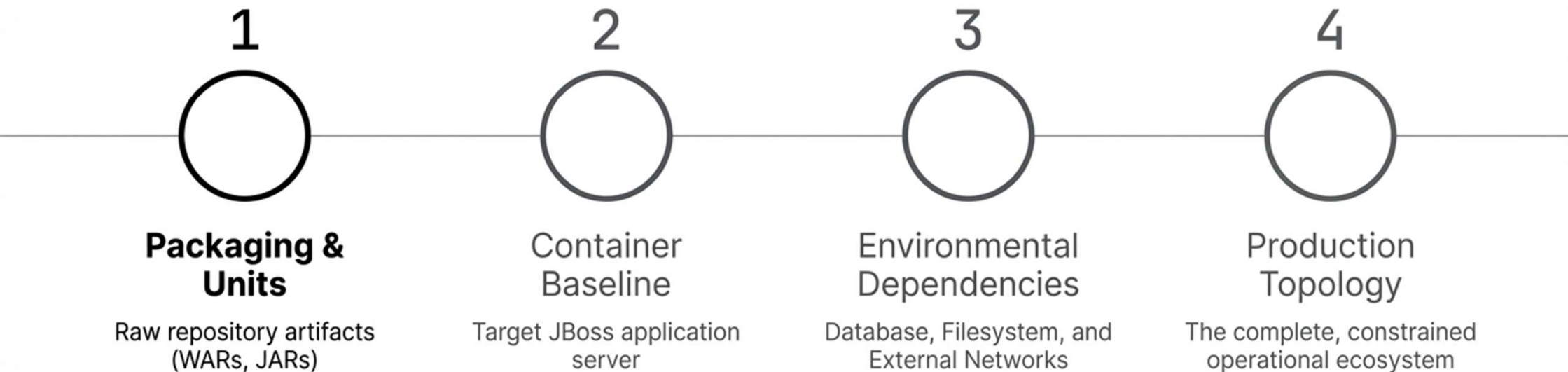


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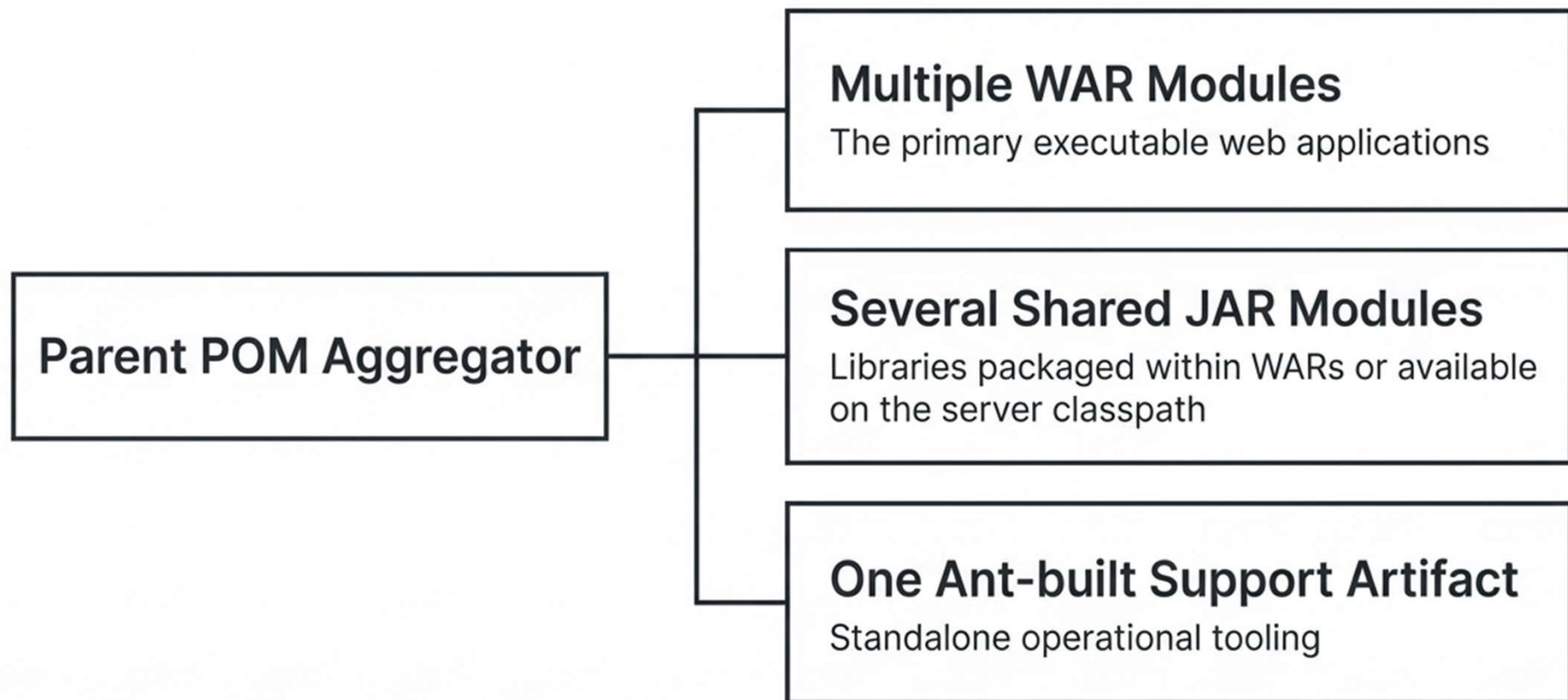
SYSTEM ANATOMY & ENVIRONMENT BASELINE

Reconstructing the deployment blueprint from raw repository artifacts

This document maps the physical and operational boundaries of the SIMOG legacy Java EE system, transforming code-level dependencies into actionable provisioning parameters.



Visible packaging structure dictates the deployable footprint.



The core runtime units consist of four distinct web archives.

Runtime Unit	Interface Type	Target Consumer	Deployment Model
SimogWeb	UI	Browser Users	Deployed to application server; dependent on shared library modules on the application classpath.
SimogWSPDD	SOAP Services	External System Integrations	Deployed to application server; dependent on shared library modules on the application classpath.
SimogWSAGG	SOAP Services	External System Integrations	Deployed to application server; dependent on shared library modules on the application classpath.
SimogWSTED	SOAP Services	External System Integrations	Deployed to application server; dependent on shared library modules on the application classpath.

EVIDENCE FOUND

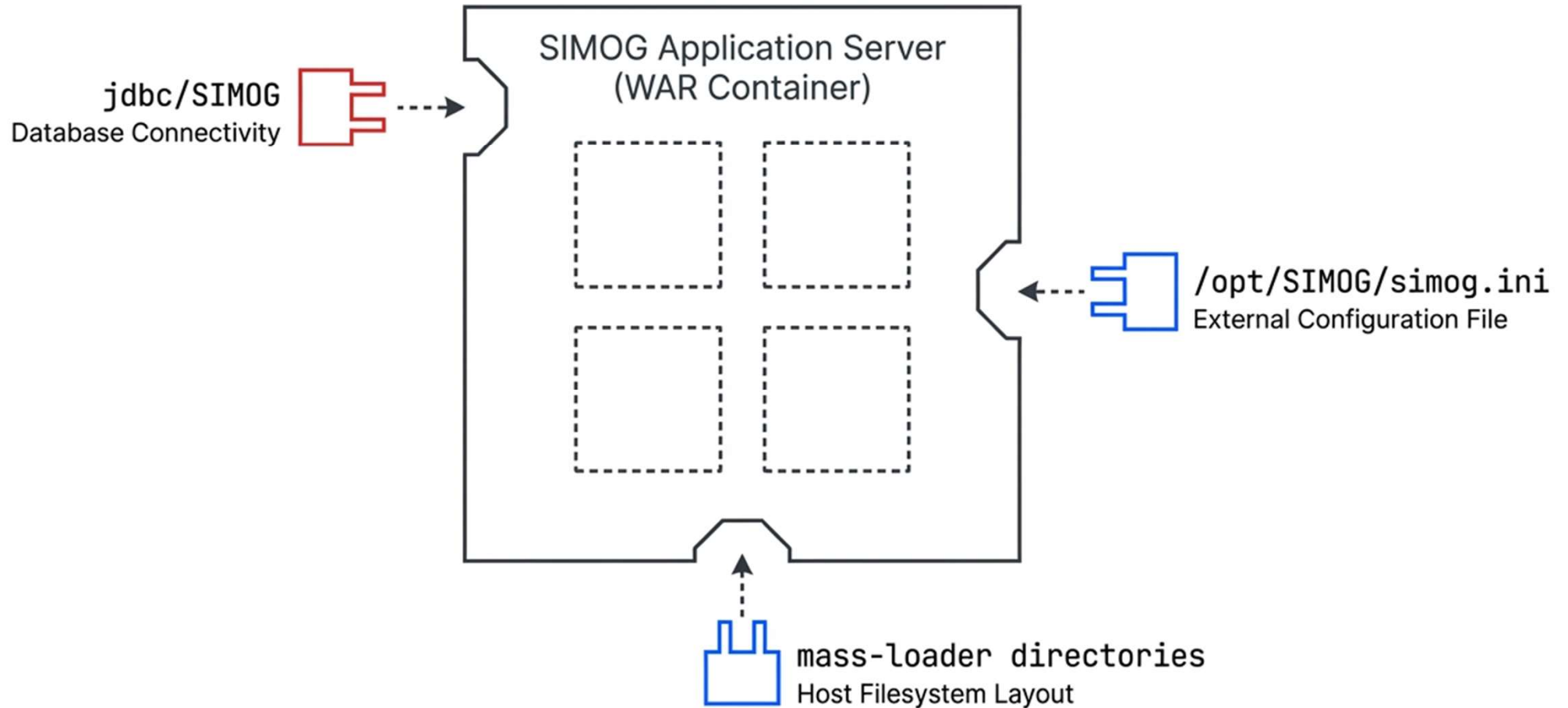
- JBoss-specific deployment descriptors
- JNDI datasource lookup patterns
- Standard Java EE deployment descriptors



OPERATIONAL INFERENCE

Target Runtime: JBoss-Family Java EE Server

Implication: The hosting environment must be a Java EE compatible container, specifically tuned for JBoss descriptor standards.



The system relies heavily on externalized configuration. Application startup will fail unless the operational environment fulfills these physical socket requirements.

Central database connectivity dictates primary runtime provisioning

PRIMARY ASSUMPTIONS

Standard Lookup:

JNDI datasource [jdbc/SIMOG](#)
or [java:/jdbc/SIMOG](#)

Alternative:

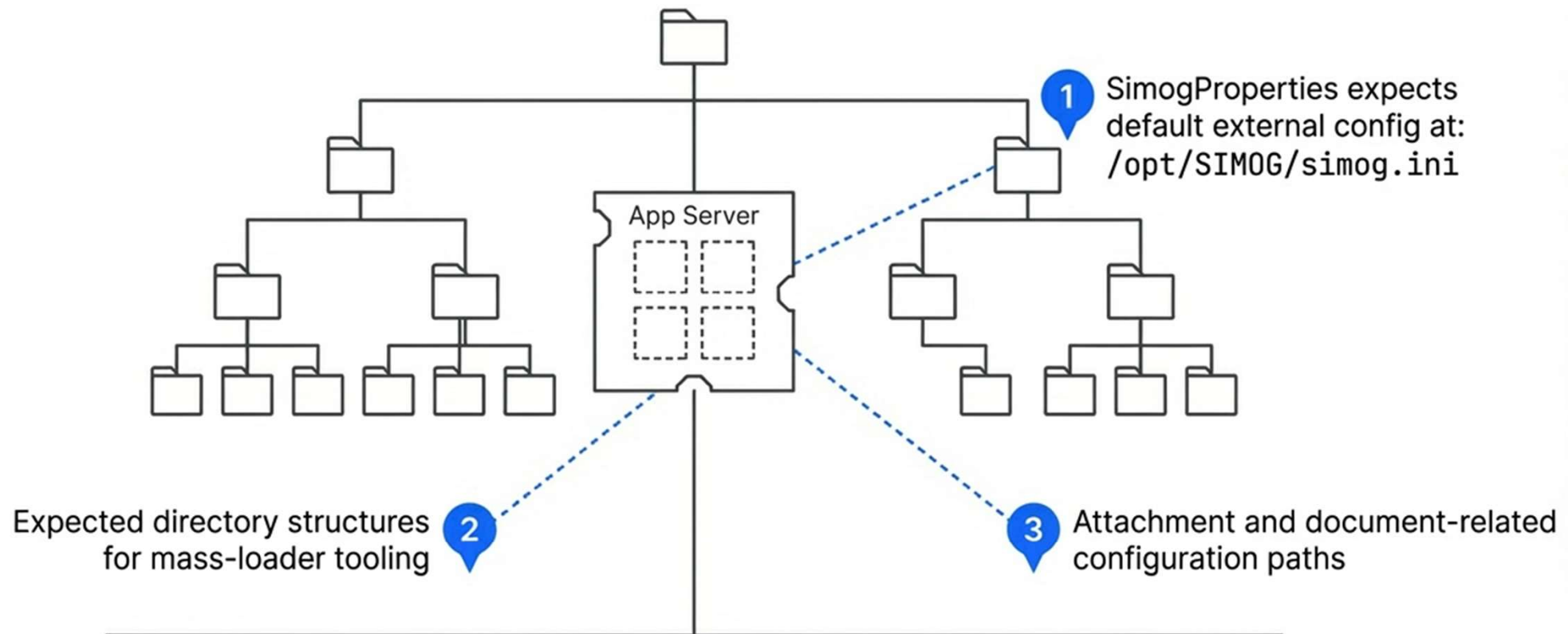
Fallback direct JDBC driver/URL support exists in specific code paths.

OPERATIONAL MANDATE

Runtime provisioning MUST explicitly include:

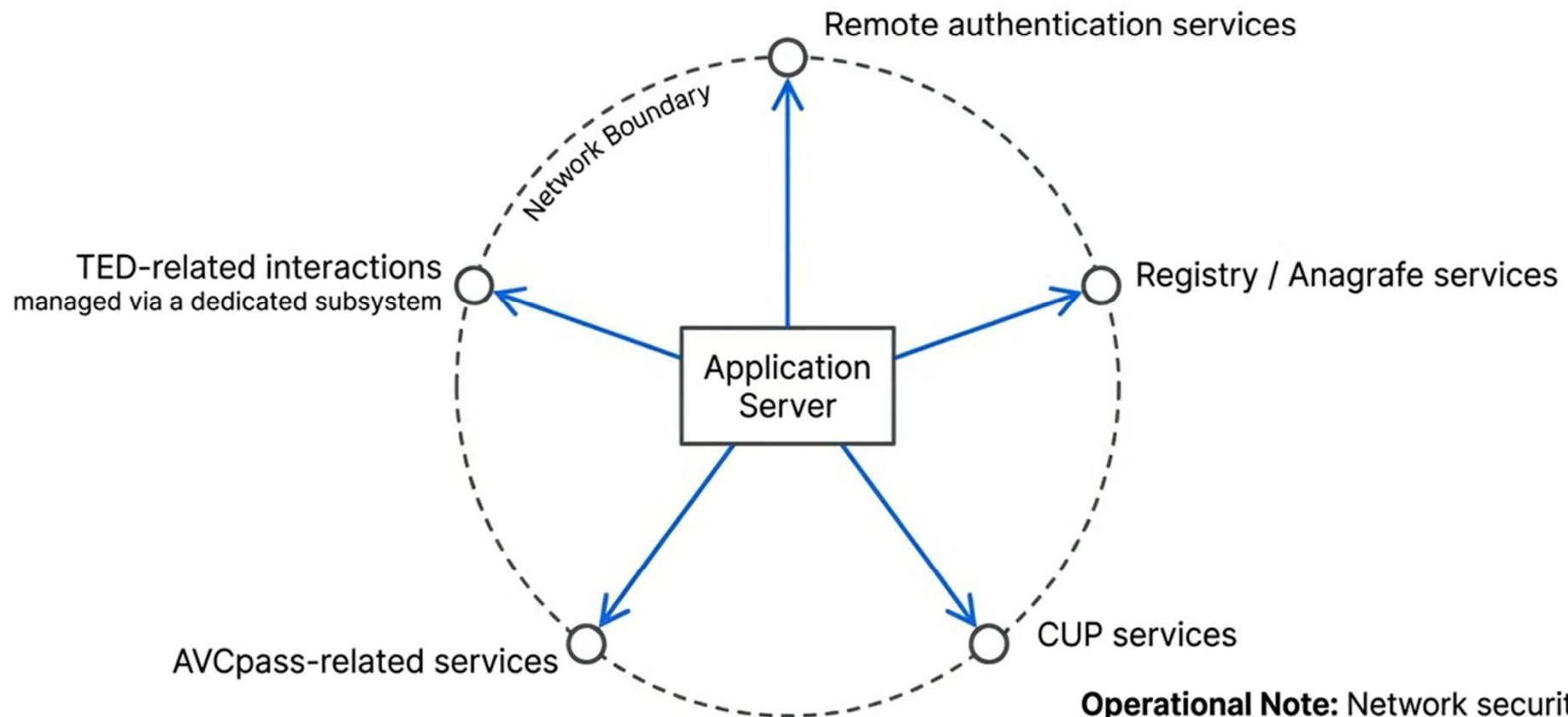
- Datasource definitions.
- Secure credential injection.
- Connection-pool tuning for shared load.

Application behavior is strictly coupled to the host filesystem layout.



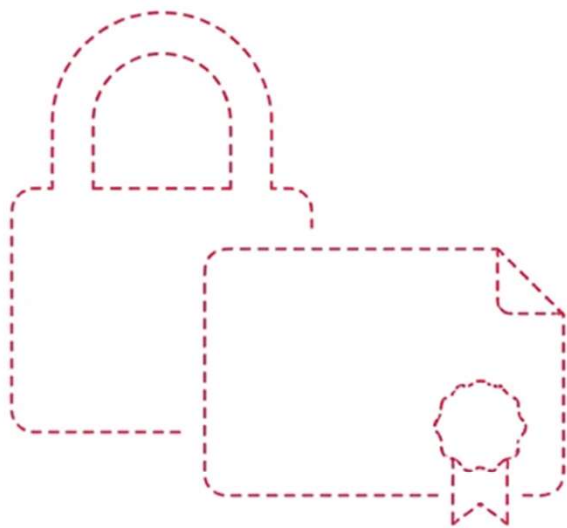
Deployments are not fully self-contained inside the WARs. The host OS environment must be pre-provisioned with specific directory structures and .ini files.

Controlled outbound network access is required for five external dependency families.



Operational Note: Network security groups must allow outbound server traffic to these specific interoperability endpoints.

Security keystores and certificates must be supplied out-of-band.



OBSERVATION

No concrete keystore artifacts were observed within the source code repository inventory.

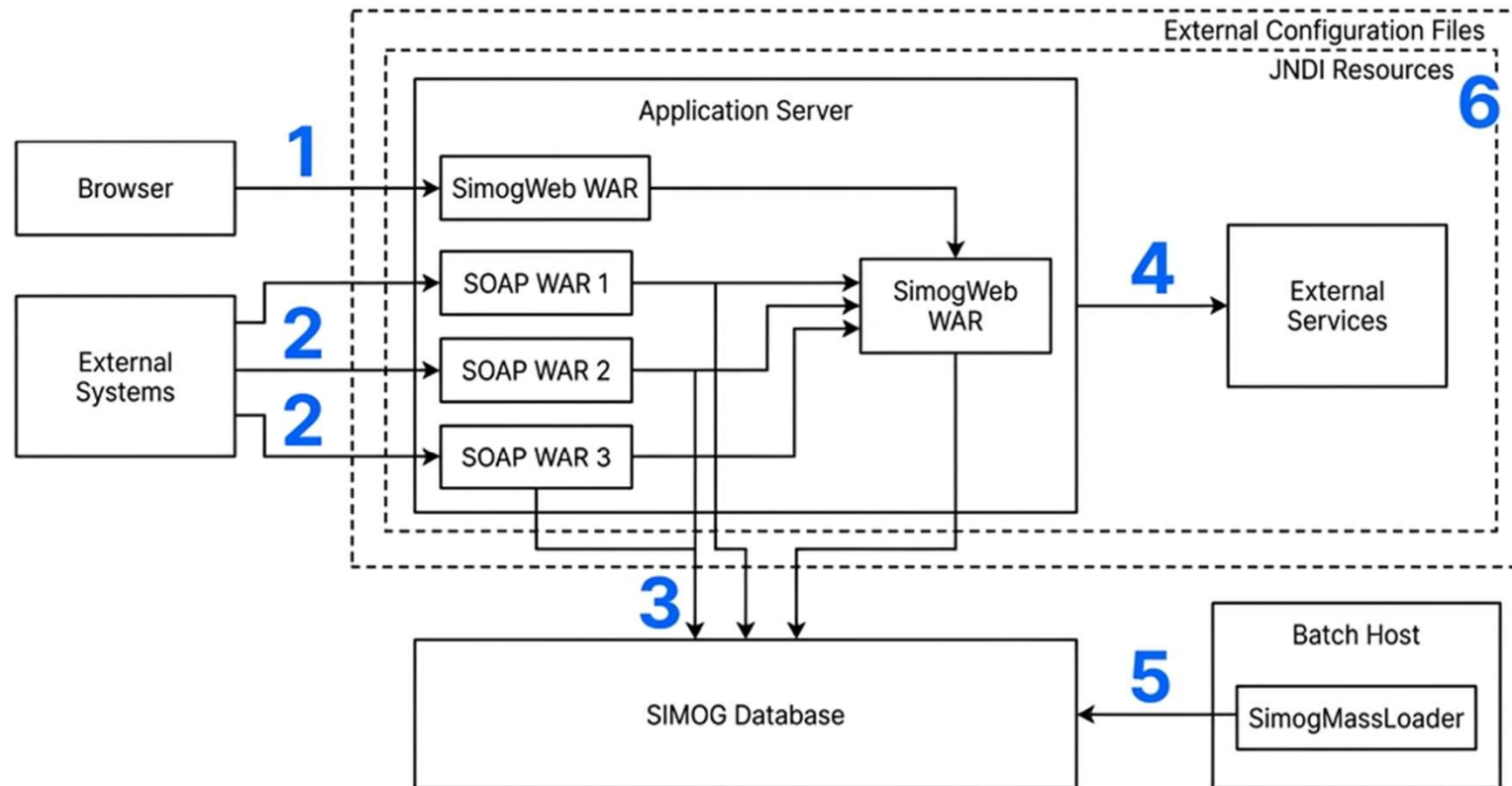
INFERENCE

The defined SOAP integrations and remote authentication services mandate transport security (TLS/SSL).

REQUIREMENT

Cryptographic material and truststores must be provisioned and configured strictly at the operational level, outside of source control.

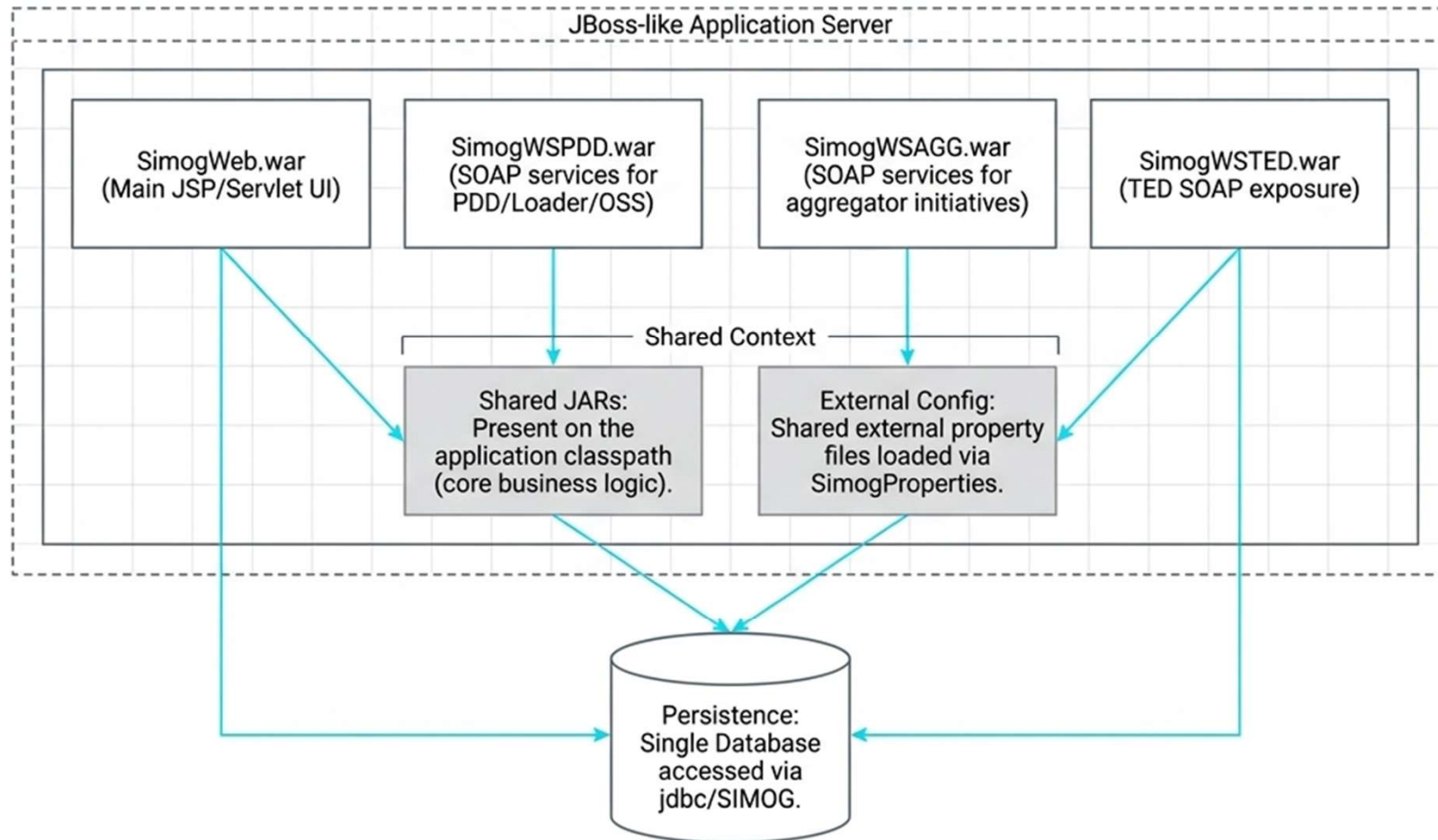
The Inferred Production Topology Blueprint



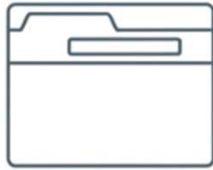
1. Browser users route network traffic to SimogWeb.
2. External systems interact with the three SOAP WARs.
3. All application modules funnel into a shared SIMOG database.

4. The application server reaches out to external authentication and registry services.
5. A batch host or scheduled OS job executes the standalone SimogMassLoader.
6. External configuration files and JNDI resources wrap the environment, shaping behavior.

Physical Topology: Containerized Deployment Model



Execution Environments: The Four Runtimes



Web Runtime

- Servlet container execution
- Intercepted by LocaleFilter & AuthenticationFilter
- ServletBase connection support -> JSP rendering



SOAP Runtime

- JAX-WS endpoint classes in dedicated WARs
- LoginActionManager & InserisciGaraActionManager
- ConnectionWSManager handles DB lifecycles



Batch Runtime

- Command-line execution (Main in SimogMassLoader)
- XML validation pipeline
- Directory-based handoff: IN, OUT, WRK, BKP



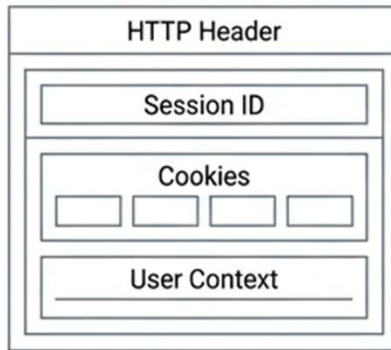
TED Runtime

- SimogWSTED SOAP endpoint
- TEDDbService backed by JPA persistence unit
- XML/XSD validation logic

Orchestration Matrix: Synchronous Execution Flows

Web Flow	Shared Database	SOAP Flow
Entry & Interception Filters process request (Locale/Auth).		Entry & Interception Endpoint receives XML/JAXB, validates mandatory inputs.
Initialization Servlet checks session via ServletBase; obtains DB connection.		Initialization Endpoint obtains DB connection; action manager validates DB Ticket.
Business Execution Dispatches by toDo parameter; invokes validators/managers.		Business Execution Manager executes domain logic; builds response bean.
Termination & Output Manual Commit/Rollback; Forward to JSP or Error Page.		Termination & Output Connection Commit/Rollback; Encode XML Technical Result.

Dual Session & Authentication Models



Web Session Security

Mechanism: Memory-backed HTTP Session.

Storage: Authenticated user context, profiles, and locale (via LocaleFilter).

Enforcement: `ServletBase.checkSession()` explicitly validates state before servlet execution. `AuthenticationFilter` enriches logs with user context (MDC).

WS_SESSIONS Table			
Ticket ID	State	Lifecycle	User ID

SOAP Session Security

Mechanism: Persistence-backed Tickets.

Storage: Physical database table `WS_SESSIONS` tracks ticket states and lifecycles.

Enforcement: WS login methods generate tickets. Action managers and `WSSessionManager` explicitly validate ticket validity before executing protected operations.